

ADVANCED QUANTUM MECHANICS

Scattering theory



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LECTURE 7: Lippmann-Schwinger theory and the Born expansion

Scattering theory provides the analytical framework for computing the rate at which particles are deflected by a target and the directions into which they are scattered. Examples include a neutron scattered by a nucleus, or an electron directed toward an atom.

The incident particle is characterized by its mass m and is in an initial state $|i\rangle$, which we pick as an eigenstate of the momentum operator $|i\rangle \equiv |\mathbf{k}_i\rangle$. The target interaction is described by a potential $V(\hat{\mathbf{r}})$. As illustrated below, we a situation where the incident particle is moving along the x -axis. The scattering process redistributes the particle into all possible directions defined by the polar and azimuthal angles (θ, ϕ) in a spherical coordinate system.

To handle the continuous momentum spectrum formally, we introduce a large quantization box of volume $L^3 = L \times L \times L$. The free-particle eigenstates are momentum eigenstates $|\mathbf{k}\rangle$ with energies $E_{\mathbf{k}} = \hbar^2 k^2 / 2m$, and, under periodic boundary conditions,

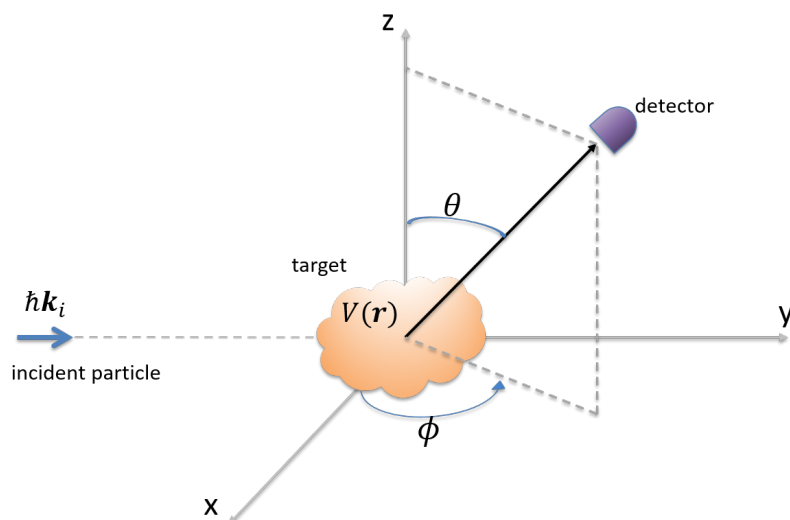
$$(k_x, k_y, k_z) = \frac{2\pi}{L}(n_x, n_y, n_z), \quad n_i \in \mathbb{Z}.$$

The problem amounts to solving the Schrödinger equation with a Hamiltonian

$$\hat{H} = \hat{H}_0 + V(\hat{\mathbf{r}}) = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{r}}),$$

where $V(\hat{\mathbf{r}})$ quantifies the effect of the target on the incident wave.

Although we ultimately wish to describe a localized wave packet, we start by considering a single plane-wave component of it. This is sufficient, because the general case follows by linear superposition. We impose asymptotic conditions corresponding to an incoming wave prepared at $t_0 = -\infty$ and evaluate the scattered wave at $t \rightarrow \infty$.



We will proceed in a slightly unconventional way: first introducing the main physical quantities used in scattering theory and outlining how they are connected. Later (in the next lecture), we will derive them systematically. For now, the goal is to build intuition for their physical meaning.

Imagine placing a small particle detector (see figure) with a small surface area ΔA at a large distance r from the target, oriented at polar angles (θ, ϕ) . The rate at which the detector records scattered particles defines the central observables of scattering theory, such as the *differential* and *total cross sections*.

Total and differential cross sections

The total scattering cross section σ_{tot} measures the effective area that quantifies how likely an incoming particle is to be scattered. It is defined as the ratio between the total number of scattered particles per unit time and the incident flux:

$$\sigma_{\text{tot}} = \frac{1}{\Phi} \frac{\Delta N_{\text{scatt}}}{\Delta t}$$

where Φ is the flux of incoming particles. Units: $[\sigma_{\text{tot}}] = \text{m}^2$. The incident flux is defined as the number of incoming particles per surface area per unit of time

$$\Phi = \frac{1}{A} \frac{\Delta N_{\text{incoming}}}{\Delta t} = \frac{1}{L^2} \frac{v}{L}$$

with dimension $[\Phi] = \text{m}^{-2} \text{s}^{-1}$. One can compute it by the number of particles in a volume element of surface L^2 times width $v\Delta t$ divided by the time element Δt . With particle density $1/L^3$, this gives the result above. To describe the angle dependence of scattering, one introduces the differential cross section as the number of particles scattered per unit in a solid angle into a solid angle element $d\Omega = \sin \theta d\theta d\phi$ as

$$\sigma_{(\theta, \phi)} = \frac{d\sigma}{d\Omega} = \frac{1}{\Phi} \frac{\Delta N_{\text{scatt}}}{\Delta t \Delta \Omega}$$

Evidently, integrating over all solid angles one recovers the total scattering cross section.

$$\sigma_{\text{tot}} = \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \frac{d\sigma}{d\Omega}$$

In order to estimate the quantities mentioned above, one proceeds in solving the Schrödinger equation to derive a matrix equation for the outgoing state vector.

Lippmann-Schwinger equation for the scattered state and the Born expansion

The Lippmann-Schwinger equation: This is an equation for the state vector at infinite time $|\psi^{(+)}\rangle$ after the scattering process

$$|\psi^{(+)}\rangle = |\mathbf{k}_0\rangle + G(E_0)\hat{V} |\psi^{(+)}\rangle \quad (1)$$

- The Green operator is defined as an inverse operator

$$\hat{G}(E_i) = \frac{1}{(E_i + i\epsilon)\mathbb{1} - \hat{H}_0}$$

where

- $E_i = \frac{(\hbar k_i)^2}{2m}$ is the initial kinetic energy (eigenvalue of $\hat{H}_0$)
- The ϵ is a small positive real number which insures that we do not divide by zero when the eigenvalue of $\hat{H}_0$ equals E_i
- The goal is to solve the equation for $|\psi^{(+)}\rangle$ – which contains all the information about the scattered directions and energies.

The Born series

Depending on the type of interaction, one can try a perturbation series (a **Born series expansion**)

$$|\psi_{\mathbf{k}_i}^{(+)}\rangle = \underbrace{|\mathbf{k}_i\rangle}_{\text{unperturbed incoming wave}} + \underbrace{\hat{G}(E_i)\hat{V}|\mathbf{k}_i\rangle}_{\text{single scattering}} + \underbrace{\hat{G}(E_i)\hat{V}\hat{G}(E_i)\hat{V}|\mathbf{k}_i\rangle}_{\text{double scattering}} + \dots \quad (2)$$

This makes sense as it describes the unperturbed wave (assuming the interaction is turned on, then the effect of a single scattering event, then of two scattering events and so on. In some cases it is enough to stop at first terms, in some other cases (as we will see in this lecture) one needs to sum up all orders of the perturbation.

We will prove these equations soon; first we want to build some intuition into what they actually mean and how one can solve them.

Building intuition - scattering off a delta potential

Let's take a simple example to build some intuition of what this theory means and can do for us. We will consider a delta potential

$$V(\mathbf{r}) = \lambda \delta^{(3)}(\mathbf{r}),$$

as a point scatterer at the origin. We will consider a finite Hilbert space spanned by momentum eigenstates

$$\{|\mathbf{k}_1\rangle, |\mathbf{k}_2\rangle, \dots, |\mathbf{k}_N\rangle\}.$$

Eventually we could take N to infinity to represent the whole continuum, but better to keep it finite now so we can understand what we do.

We consider an incoming plane wave, so a state described by the ket $|\mathbf{k}_i\rangle$ (with i chosen from the numbers 1 to N). We choose to represent any initial state by a properly placed unity in a column vector with N entries, at position i :

$$|\mathbf{k}_i\rangle = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{entry } i \leftarrow$$

This means all other plane waves are initially empty. Assuming no potential, of course the final state would be the same as the incoming one and all other plane waves will still be empty. However, let us now write the final state after all possible scattering processes

$$|\psi_{\mathbf{k}_i}^{(+)}\rangle = \underbrace{|\mathbf{k}_i\rangle}_{\text{unperturbed incoming wave}} + \underbrace{\hat{G}(E_i)V|\mathbf{k}_i\rangle}_{\text{single scattering}} + \underbrace{\hat{G}(E_i)V\hat{G}(E_i)V|\mathbf{k}_i\rangle}_{\text{double scattering}} + \dots \quad (3)$$

The zeroth order is obvious as it is the unscattered component. To compute the next ones, let us express the matrix forms for $\hat{V}$ and $\hat{G}(E_i)$ in the momentum basis. For the potential we have

$$V_{jj'} = \langle \mathbf{k}_j | \hat{V} | \mathbf{k}_{j'} \rangle = \lambda \int d^3r \langle \mathbf{k}_j | \mathbf{r} \rangle \delta^{(3)}(\mathbf{r}) \langle \mathbf{r} | \mathbf{k}_{j'} \rangle$$

with

$$\langle \mathbf{r} | \mathbf{k}'_j \rangle = \frac{1}{\sqrt{L^3}} e^{i\mathbf{k}'_j \cdot \mathbf{r}}, \quad \langle \mathbf{k}_j | \mathbf{r} \rangle = \frac{1}{\sqrt{L^3}} e^{-i\mathbf{k}_j \cdot \mathbf{r}}.$$

Thus

$$V_{jj'} = \lambda \frac{1}{L^3} \int d^3r e^{-i\mathbf{k}_j \cdot \mathbf{r}} \delta^{(3)}(\mathbf{r}) e^{i\mathbf{k}_{j'} \cdot \mathbf{r}} = \lambda \frac{1}{L^3} e^{i(\mathbf{k}_{j'} - \mathbf{k}_j) \cdot \mathbf{0}} = \frac{\lambda}{L^3}.$$

Therefore, in the finite $\mathbf{k}$ -basis the potential matrix has constant entries,

$$V = \frac{\lambda}{L^3} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}.$$

This allows to easily see the first application

$$V |\mathbf{k}_0\rangle = \frac{\lambda}{L^3} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \frac{\lambda}{L^3} \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

The free Green operator (resolvent) is defined as

$$\hat{G}(E_i) = \frac{1}{E_i \mathbb{1} - \hat{H}_0 + i\epsilon \mathbb{1}}.$$

In the momentum basis $\{|\mathbf{k}_i\rangle\}$, the kinetic Hamiltonian is diagonal, therefore, the matrix in the denominator is also diagonal:

$$E_i \mathbb{1} - \hat{H}_0 + i\epsilon \mathbb{1} = \begin{pmatrix} E_i - E_{\mathbf{k}_1} + i\epsilon & 0 & \cdots & 0 \\ 0 & E_i - E_{\mathbf{k}_2} + i\epsilon & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & E_i - E_{\mathbf{k}_N} + i\epsilon \end{pmatrix}$$

This makes taking the inverse easy, and we compute that

$$\hat{G}(E_i) = \begin{pmatrix} u_1 & 0 & \cdots & 0 \\ 0 & u_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_N \end{pmatrix}, \quad u_j = \frac{1}{E_i - E_{\mathbf{k}_j} + i\epsilon}.$$

We can put it all together to show the first order of scattering to be

$$\hat{G}(E_i)V = \frac{\lambda}{L^3} \begin{pmatrix} u_1 & 0 & \cdots & 0 \\ 0 & u_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_N \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix} = \frac{\lambda}{L^3} \begin{pmatrix} u_1 & u_1 & \cdots & u_1 \\ u_2 & u_2 & \cdots & u_2 \\ \vdots & \vdots & \ddots & \vdots \\ u_N & u_N & \cdots & u_N \end{pmatrix},$$

and

$$\hat{G}(E_i)V |\mathbf{k}_i\rangle = \frac{\lambda}{L^3} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{pmatrix}.$$

So the single-scattering term distributes the amplitude across all momentum components with weights u_j . Up to this order, we can write

$$|\psi_{\mathbf{k}_i}^{(+)}\rangle = \begin{pmatrix} \psi_1^{(+)} \\ \psi_2^{(+)} \\ \vdots \\ \psi_N^{(+)} \end{pmatrix} = \begin{pmatrix} \delta_{1,i} + \frac{\lambda}{L^3} u_1 \\ \delta_{2,i} + \frac{\lambda}{L^3} u_2 \\ \vdots \\ \delta_{N,i} + \frac{\lambda}{L^3} u_N \end{pmatrix} = \begin{pmatrix} \delta_{1,i} + \frac{\lambda}{L^3} \frac{1}{E_i - E_{\mathbf{k}_1} + i\epsilon} \\ \delta_{2,i} + \frac{\lambda}{L^3} \frac{1}{E_i - E_{\mathbf{k}_2} + i\epsilon} \\ \vdots \\ \delta_{N,i} + \frac{\lambda}{L^3} \frac{1}{E_i - E_{\mathbf{k}_N} + i\epsilon} \end{pmatrix}.$$

Notice now that

$$\hat{G}(E_i) V \hat{G}(E_i) V |\mathbf{k}_i\rangle = \left(\frac{\lambda}{L^3}\right)^2 \begin{pmatrix} u_1 & u_1 & \cdots & u_1 \\ u_2 & u_2 & \cdots & u_2 \\ \vdots & \vdots & \ddots & \vdots \\ u_N & u_N & \cdots & u_N \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{pmatrix} = \left(\frac{\lambda}{L^3}\right)^2 \left(\sum_{\ell=1}^N u_\ell\right) \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{pmatrix}.$$

We can recognize the following pattern for the j -th component:

$$\psi_j^{(+)} = \delta_{j,i} + u_j \left[\frac{\lambda}{L^3} + \left(\frac{\lambda}{L^3}\right)^2 \sum_{\ell=1}^N u_\ell + \left(\frac{\lambda}{L^3}\right)^3 \left(\sum_{\ell=1}^N u_\ell\right)^2 + \cdots \right].$$

Let $\alpha \equiv \frac{\lambda}{L^3} \sum_{\ell=1}^N u_\ell$. The bracket is a geometric series which can be summed up to:

$$\frac{\lambda}{L^3} (1 + \alpha + \alpha^2 + \cdots) = \frac{\lambda/L^3}{1 - \alpha} = \frac{\lambda/L^3}{1 - \frac{\lambda}{L^3} \sum_{\ell=1}^N u_\ell}.$$

Therefore,

$$\psi_j^{(+)} = \delta_{j,i} + \frac{\lambda/L^3}{1 - \frac{\lambda}{L^3} \sum_{\ell=1}^N u_\ell} u_j$$

We have found what the amplitudes of the different states are. To proceed, we will simplify our problem, as it is more complicated in 3 dimensions, where some renormalization procedure is required.

Building more intuition - scattering off a delta potential in one dimension

Let us now narrow down to an easier problem, by considering a 1D problem to apply the Lippmann–Schwinger theory. We assume the same problem as before but reduced to one dimension, so

$$\hat{H} = \frac{\hat{p}^2}{2m} + \lambda \delta(\hat{x}), \quad \hat{V}(x) = \lambda \delta(x),$$

The matrix elements of the potential are

$$V_{kk'} = \langle k | V | k' \rangle = \frac{\lambda}{L} \int dx e^{-ikx} \delta(x) e^{ik'x} = \frac{\lambda}{L},$$

Iterating as before but in 1D, we obtain a Born series

$$|\psi^{(+)}\rangle = |k_i\rangle + \hat{G}(E_i) \hat{V} |k_i\rangle + \hat{G}(E_i) \hat{V} \hat{G}(E_i) \hat{V} |k_i\rangle + \cdots.$$

Summing as before, we find that the j -th component satisfies

$$\psi_j^{(+)} = \delta_{ji} + \frac{\lambda/L}{1 - \frac{\lambda}{L} \sum_{\ell} u_{\ell}} u_j, \quad u_{\ell} = \frac{1}{E_i - E_{\ell} + i\epsilon}.$$

Let us now evaluate the sum by moving to the continuum. First let's detail what the transition to continuum means. The periodic boundary condition that the function repeats itself after L is $\psi(x + L) = \psi(x)$ which for a plane wave $\psi_k(x) = e^{ikx}$ is satisfied this only if

$$k = \frac{2\pi n}{L}, \quad n \in \mathbb{Z}.$$

Let us detail below how to think about the transition from discrete to continuum.

Transition to the continuum - Density of states

Fix an interval in k -space, $k_1 = \Delta k n_1$ and $k_2 = \Delta k n_2$, with integers $n_1 < n_2$. We will ask that for any function f the equality holds:

$$\sum_{n=n_1}^{n_2} f(k_n) = \int_{k_1}^{k_2} dk \rho(k) f(k).$$

Discrete picture: The number of states in this interval is

$$n_{\text{states}} = \sum_{n=n_1}^{n_2} 1 = n_2 - n_1.$$

Continuum picture: The number of states in the same interval calculated in the continuum is

$$n_{\text{states}} = \int_{k_1}^{k_2} \rho(k) dk,$$

where the probability distribution $\rho(k)$ is to be derived (and will end up being the density of states). Equating the two

$$\int_{k_1}^{k_2} \rho(k) dk = \frac{k_2 - k_1}{\Delta k}.$$

Assuming $\rho(k)$ constant inside the interval,

$$\rho(k) (k_2 - k_1) = \frac{k_2 - k_1}{\Delta k} \Rightarrow \rho(k) = \frac{1}{\Delta k} = \frac{L}{2\pi}.$$

Therefore, the sum over discrete momenta over the whole interval becomes

$$\sum_n f(k_n) = \frac{L}{2\pi} \int_{-\infty}^{\infty} dk f(k).$$

Now we are in the position of computing the sum appearing in the denominator by transitioning to the continuum

$$\frac{1}{L} \sum_{\ell} u_{\ell} = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dk \frac{1}{E_i - \frac{\hbar^2 k^2}{2m} + i\epsilon} = \frac{2m}{2\pi \hbar^2} \int_{-\infty}^{+\infty} dk \frac{1}{k_i^2 - k^2 + i\epsilon'}$$

where we renormalized the ϵ to ϵ' . This is where finally we see the reason for introducing the small positive parameter ϵ , which allows for the integral to converge (using contour integration - explained in detail in the Appendix)

$$\int_{-\infty}^{+\infty} dk \frac{1}{k_i^2 - k^2 + i\epsilon'} = -\frac{i\pi}{k_i}.$$

Putting it all together

$$\frac{1}{L} \sum_{\ell} u_{\ell} = -\frac{im}{\hbar^2 k_i}.$$

We can now write the wavefunction by projecting into the position space

$$\psi^{(+)}(x) = \langle x | \psi^{(+)} \rangle = \frac{1}{\sqrt{L}} \sum_j \psi_j^{(+)} e^{ik_j x}.$$

In the continuum limit,

$$\psi^{(+)}(x) = \frac{1}{\sqrt{2\pi}} \left[e^{ik_i x} + \frac{\lambda/L}{1 + i \frac{m\lambda}{\hbar^2 k_i}} \int_{-\infty}^{+\infty} dk \frac{e^{ikx}}{E_i - E_k + i\epsilon} \right].$$

This is similar to the procedure detailed in the Appendix which we used to evaluate the previous integral. By contour integration, the result is

$$\int_{-\infty}^{+\infty} dk \frac{e^{ikx}}{k_i^2 - k^2 + i\epsilon} = -\frac{i\pi}{k_i} e^{ik_i |x|}.$$

The factor $e^{ik_i |x|}$ equals

$$e^{ik_i x} \text{ for } x > 0, \quad e^{-ik_i x} \text{ for } x < 0.$$

Therefore the δ -potential scatters an incoming $e^{ik_i x}$ only into $e^{ik_i x}$ (transmitted) and $e^{-ik_i x}$ (reflected) components.

Matching to the standard scattering form

$$\psi(x < 0) = e^{ik_i x} + r e^{-ik_i x}, \quad \psi(x > 0) = t e^{ik_i x},$$

we obtain

$$r = \frac{-i \frac{m\lambda}{\hbar^2 k_i}}{1 + i \frac{m\lambda}{\hbar^2 k_i}}, \quad t = \frac{1}{1 + i \frac{m\lambda}{\hbar^2 k_i}}.$$

This is of course the same result as we obtained by solving the stationary Schrödinger equation in Exercise 3 as

$$t = \frac{1}{1 + i\eta}, \quad r = \frac{-i\eta}{1 + i\eta}, \quad t = 1 + r$$

with $\eta = \frac{m\lambda}{\hbar^2 k}$.

Appendix: Computing integrals via residue theorem

We want to compute

$$I \equiv \int_{-\infty}^{+\infty} dk \frac{1}{k_i^2 - k^2 + i\epsilon'}$$

with $k_i > 0$ and $\epsilon' > 0$ infinitesimal. Claim:

$$I = -\frac{i\pi}{k_i}.$$

1. Locate the poles

Define

$$f(k) = \frac{1}{k_i^2 - k^2 + i\epsilon'}.$$

Poles are where the denominator vanishes:

$$k_i^2 - k^2 + i\epsilon' = 0 \quad \Rightarrow \quad k^2 = k_i^2 + i\epsilon'.$$

Call the two roots

$$k_{\pm} = \pm \sqrt{k_i^2 + i\epsilon'}.$$

For small $\epsilon' > 0$, expand

$$\sqrt{k_i^2 + i\epsilon'} = k_i \sqrt{1 + i\epsilon'/k_i^2} \approx k_i \left(1 + \frac{i\epsilon'}{2k_i^2}\right) = k_i + i\frac{\epsilon'}{2k_i}.$$

Thus approximately:

$$k_+ \approx k_i + i\frac{\epsilon'}{2k_i} \quad (\text{slightly above real axis}),$$

$$k_- \approx -k_i - i\frac{\epsilon'}{2k_i} \quad (\text{slightly below real axis}).$$

Thus: - one pole in the **upper half-plane** at k_+ , - one pole in the **lower half-plane** at k_- .

2. Choose the contour

Consider the contour integral of $f(k)$ along: - the real axis from $-R$ to R , - plus a semicircle of radius R in the **upper half-plane**.

Call the full contour C_R . Then by Cauchy's theorem:

$$\oint_{C_R} dk f(k) = 2\pi i \sum (\text{residues inside } C_R).$$

Only the pole k_+ lies inside the upper half-plane contour:

$$\oint_{C_R} dk f(k) = 2\pi i \text{Res}(f, k_+).$$

Split the contour:

$$\oint_{C_R} dk f(k) = \int_{-R}^R dk f(k) + \int_{\text{arc}} dk f(k).$$

We send $R \rightarrow \infty$ and show the arc contribution vanishes.

3. Vanishing of the arc

On the arc, $k = Re^{i\theta}$ with $\theta \in [0, \pi]$. For large R :

$$f(k) = \frac{1}{k_i^2 - k^2 + i\epsilon'} \sim \frac{1}{-k^2} \sim \frac{1}{R^2}.$$

The length of the arc is πR , so

$$\left| \int_{\text{arc}} dk f(k) \right| \lesssim (\pi R) \cdot \frac{1}{R^2} = \frac{\pi}{R} \xrightarrow{R \rightarrow \infty} 0.$$

Hence, in the limit $R \rightarrow \infty$:

$$\int_{-\infty}^{+\infty} dk f(k) = 2\pi i \operatorname{Res}(f, k_+).$$

Thus the integral we want is $2\pi i$ times the residue at the upper pole.

4. Compute the residue at k_+

For a simple pole at k_0 ,

$$\operatorname{Res}(f, k_0) = \left. \frac{1}{\frac{d}{dk}(k_i^2 - k^2 + i\epsilon')} \right|_{k=k_0}.$$

Compute the derivative:

$$\frac{d}{dk}(k_i^2 - k^2 + i\epsilon') = -2k.$$

So at $k = k_+$:

$$\operatorname{Res}(f, k_+) = \frac{1}{-2k_+} = -\frac{1}{2k_+}.$$

In the limit $\epsilon' \rightarrow 0^+$, $k_+ \rightarrow k_i$, so

$$\operatorname{Res}(f, k_+) \longrightarrow -\frac{1}{2k_i}.$$

5. Final result

$$I = \int_{-\infty}^{+\infty} dk f(k) = 2\pi i \operatorname{Res}(f, k_+) = 2\pi i \left(-\frac{1}{2k_i} \right) = -\frac{i\pi}{k_i}.$$